EXPERIMENT – 21 :Towers of Hanoi Using Prolog

Aim:

To write and execute a Prolog program to solve the Towers of Hanoi problem using recursion.

Software Required:

- SWI-Prolog
- Windows Operating System

Program

File Name: exp21.pl

File Path: C:/ai prolog/exp21.pl

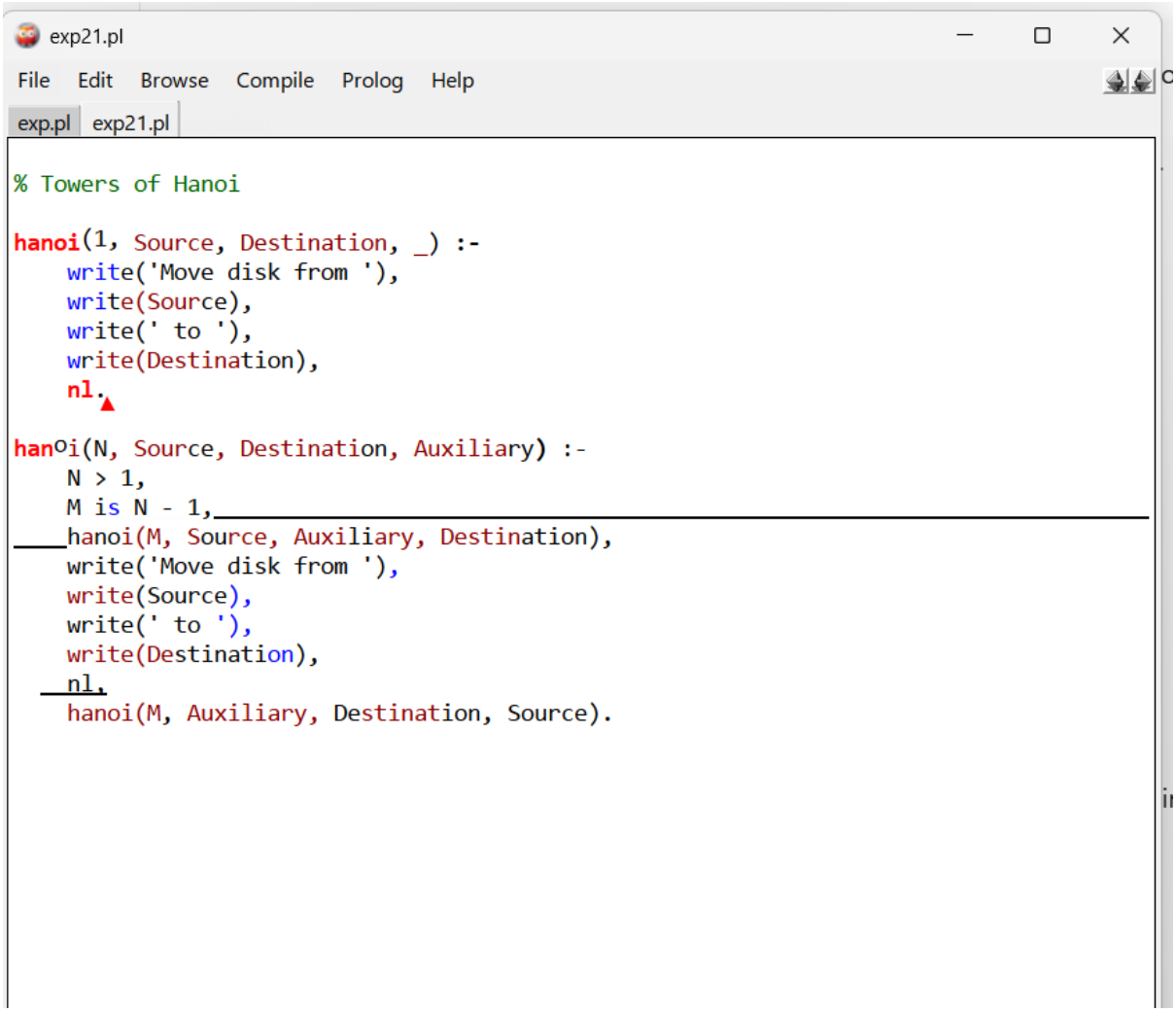
```
% Towers of Hanoi

hanoi(1, Source, Destination, _) :-
    write('Move disk from '),
    write(Source),
    write(' to '),
    write(Destination),
    nl.

hanoi(N, Source, Destination, Auxiliary) :-
    N > 1,
    M is N - 1,
    hanoi(M, Source, Auxiliary, Destination),
    write('Move disk from '),
    write(Source),
    write(' to '),
    write(Destination),
```

nl,

hanoi(M, Auxiliary, Destination, Source).



Procedure

1. Open SWI-Prolog software.
2. Create a new file and type the above Prolog program.
3. Save the file as exp21.pl in the C:/ai prolog/ folder.
4. Load the program using the following command:

?- consult('C:/ai prolog/exp21.pl').

5. Execute the query:

?- hanoi(3, a, c, b).

6. Observe the output displayed in the Prolog console.

Output

```
39 ?- consult('c:/ai prolog/exp21.pl').
true.

40 ?- hanoi( 3, a, c, b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
```

Explanation

The Towers of Hanoi problem consists of three rods:

- A: Source rod
- B: Auxiliary rod
- C: Destination rod

The objective is to move all disks from the source rod to the destination rod by following these rules:

1. Only one disk can be moved at a time.
2. Only the top disk can be moved.
3. A larger disk cannot be placed on a smaller disk.

The program uses recursion to move the disks from one rod to another.

Algorithm

1. If the number of disks is 1, move the disk directly to the destination.
2. Move N−1 disks from the source rod to the auxiliary rod.
3. Move the largest disk from the source rod to the destination rod.
4. Move N−1 disks from the auxiliary rod to the destination rod.

Formula

Minimum number of moves:

\[M=2^N-1 \]

For 3 disks:

\[M=2^3-1=7 \]

Result

Thus, the Prolog program for solving the Towers of Hanoi problem using recursion was written and executed successfully.